SST 412: DESIGN AND ANALYSIS OF EXPERIMENTS

Chapter Notes — Completely Randomized Design, Randomized Block Design, Latin Squares, BIBD and Design Comparison

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These notes are the extracted content of the SST 412 chapter notes (78 pages in the source file, of which pages 1–64 carry text and pages 65–78 are blank). The prose was taken from the text layer of the document and re-set here; the mathematical working was rebuilt from the same text layer, in which formulas lose their two-dimensional layout, so the algebra that the source sets line-by-line has been restored to standard form and long expansions have been condensed to the result they establish, with the source page noted where that has been done.

Chapter map: 1 — completely randomized design (fixed and random effects); 2 — randomized block design (fixed and random effects); 3 — estimation of a missing observation in a randomized block design; 4 — Latin square design; 5 — Graeco–Latin squares; 6 — balanced incomplete block design; 7 — comparison of designs. The source material ends inside Chapter 7, at the comparison of the Latin square design with the completely randomized design.

CHAPTER 1: COMPLETELY RANDOMIZED DESIGN

I. CRD — Fixed Effect Model

In this field plan the field is first divided into 'n' equal-sized plots. All the plots are assumed to be uniform in every way. The 'v' treatments are then assigned to these n plots in a random order.

Let the ith treatment appear in n_i plots, $i = 1, 2, \ldots, v$. Further, let y_{ij} denote the jth observation of the ith treatment, $j = 1, 2, \ldots, n_i$. y_{ij} is then a random variable. We shall assume that

$$E(y_{ij}) = \mu + t_i, \qquad \text{var}(y_{ij}) = \sigma^2 \text{ (unknown constant)}, \qquad \text{cov}(y_{ij}, y_{i'j'}) = 0, \quad i \neq i',\ j \neq j'$$

where μ is some unknown constant and t_i is some unknown constant measuring the effect of the ith treatment.

Such a model is known as a **linear fixed-effect additive model**. We wish to test

$$H_0: t_1 = t_2 = \cdots = t_v = 0$$

against the alternative $H_a: t_i \neq 0$ for some i.

Analysis Procedure

Step I. We shall obtain the best unbiased estimates of $\mu, t_1, t_2, \ldots, t_v$ and σ^2. With the above assumptions the best method is the method of least squares (L.S.), which gives the minimum-variance unbiased estimators.

Description of the method

Let

$$S = \sum_{i=1}^{v} \sum_{j=1}^{n_i} \left(y_{ij} - E(y_{ij})\right)^2 = \sum_{i=1}^{v} \sum_{j=1}^{n_i} \left(y_{ij} - \mu - t_i\right)^2$$

Choose μ and t_i such that S is minimum. The conditions of minimum are:

1. $\frac{\partial S}{\partial \mu} = 0$
2. $\frac{\partial S}{\partial t_i} = 0, i = 1, 2, \ldots, v$

These two conditions give rise to $v + 1$ equations:

1. gives $\sum_{i=1}^{v} \sum_{j=1}^{n_i} -2\left(y_{ij} - \mu - t_i\right) = 0$
2. gives $\sum_{j=1}^{n_i} -2\left(y_{ij} - \mu - t_i\right) = 0$

We note that summing equation 2) over i gives equation 1). This implies that the number of independent equations is $(v + 1) - 1$, i.e. we have $v + 1$ unknowns to estimate from v equations. Provided that the equations are solvable, the number of solutions is infinite.

For uniqueness of solution take one additional equation independent of 1) and 2). Consider the additional equation, say,

$$\sum_{i=1}^{v} t_i = 0.$$

Note that whatever additional equation we take the solution will change but the analysis remains invariant.

From equation 1) we obtain

$$\sum_{i=1}^{v} \sum_{j=1}^{n_i} y_{ij} = \mu \sum_{i=1}^{v} \sum_{j=1}^{n_i} 1 + \sum_{i=1}^{v} \sum_{j=1}^{n_i} t_i$$

which gives

$$\hat{\mu} = \bar{y}_{..} = \frac{1}{n} \sum_{i=1}^{v} \sum_{j=1}^{n_i} y_{ij}$$

where $n = \sum_{i=1}^{v} n_i$, the total number of plots in the field.

Equation 2) gives

$$\sum_{j=1}^{n_i} y_{ij} = n_i\mu + n_i t_i \qquad \Rightarrow \qquad \frac{1}{n_i}\sum_{j=1}^{n_i} y_{ij} = \mu + t_i$$

so that

$$\hat{t}_i = \bar{y}_{i.} - \bar{y}_{..}$$

Going back to the model we have

$$E(y_{ij}) = \mu + t_i, \qquad \text{var}(y_{ij}) = \sigma^2, \qquad \text{cov}(y_{ij}, y_{i'j'}) = 0, \quad i \neq i', j \neq j'$$

To test $H_0: t_1 = t_2 = \cdots = t_v = 0$ against $H_a: t_i \neq 0$ for some i, we further assume that

$$y_{ij} \sim N\Big[E(y_{ij}), \sigma^2\Big].$$

We write the model as

$$y_{ij} = \mu + t_i + e_{ij} \qquad \text{where } e_{ij} \sim N\Big[0, \sigma^2\Big]$$

This is equivalent to $y_{ij} \sim N\Big[E(y_{ij}), \sigma^2\Big]$; e_{ij} is called the **error variable**.

Substituting $\hat{\mu}$ and $\hat{t}_i = \bar{y}_{i.} - \bar{y}_{..}$ in the model we get

$$y_{ij} = \bar{y}_{..} + \left(\bar{y}_{i.} - \bar{y}_{..}\right) + \left(y_{ij} - \bar{y}_{i.}\right)$$

or

$$y_{ij} - \bar{y}_{..} = \left(\bar{y}_{i.} - \bar{y}_{..}\right) + \left(y_{ij} - \bar{y}_{i.}\right)$$

Squaring and summing both sides we get

$$\sum_{i=1}^{v}\sum_{j=1}^{n_i}\left(y_{ij} - \bar{y}_{..}\right)^2 = \sum_{i=1}^{v}\sum_{j=1}^{n_i}\left(\bar{y}_{i.} - \bar{y}_{..}\right)^2 + \sum_{i=1}^{v}\sum_{j=1}^{n_i}\left(y_{ij} - \bar{y}_{i.}\right)^2 + 2\sum_{i=1}^{v}\sum_{j=1}^{n_i}\left(\bar{y}_{i.} - \bar{y}_{..}\right)\left(y_{ij} - \bar{y}_{i.}\right)$$

The third term vanishes. Hence

$$\sum_{i=1}^{v}\sum_{j=1}^{n_i}\left(y_{ij} - \bar{y}_{..}\right)^2 = \sum_{i=1}^{v} n_i\left(\bar{y}_{i.} - \bar{y}_{..}\right)^2 + \sum_{i=1}^{v}\sum_{j=1}^{n_i}\left(y_{ij} - \bar{y}_{i.}\right)^2$$

$$\text{i.e.} \qquad (TSS) = (SS\,TMT) + (SSE)$$

Thus we can form the analysis of variance table as follows.

ANOVA Table

Source of variation	df	SS	MSS	F
Treatment	$v-1$	$\sum_{i=1}^{v} n_i\left(\bar{y}_{i.} - \bar{y}_{..}\right)^2$	$T = \frac{SS\,TMT}{v-1}$	$\frac{T}{E}$
Error	$n-v$	$\sum_{i=1}^{v}\sum_{j=1}^{n_i}\left(y_{ij} - \bar{y}_{i.}\right)^2$	$E = \frac{SSE}{n-v}$	
Total	$n-1$	$\sum_{i=1}^{v}\sum_{j=1}^{n_i}\left(y_{ij} - \bar{y}_{..}\right)^2$		

To test the above hypothesis compute

$$F_c = \frac{T}{E} = \frac{MS\,TMT}{MS\,E}$$

and reject $H_0: t_1 = t_2 = \cdots = t_v = 0$ at the α level whenever $F_c > F_\alpha(n_1, n_2)$, the tabular value.

Note that when $H_0: t_1 = \cdots = t_v = 0$ is true,

$$F_c = \frac{T}{E} \sim F\text{-distribution with } (n_1, n_2) \text{ degrees of freedom.}$$

Conclusion

1. If H_0 is rejected then all t_i's are not equal, i.e. some t_i's are preferable to the rest.
2. If H_0 is accepted then there is no choice between t_i's.

It can be shown that if

1. $U = \frac{1}{\sigma^2}\sum_{i=1}^{v} n_i\left(\bar{y}_{i.} - \bar{y}_{..}\right)^2 \sim \chi^2$ with $v-1$ degrees of freedom when $H_0: t_1 = \cdots = t_v = 0$ is true,
2. $W = \frac{1}{\sigma^2}\sum_{i=1}^{v}\sum_{j=1}^{n_i}\left(y_{ij} - \bar{y}_{i.}\right)^2 \sim \chi^2$ with $n-v$ degrees of freedom,
3. both U and W are stochastically independent,

then it follows that

$$F = \frac{U/(v-1)}{W/(n-v)} = \frac{T}{E} \sim F(n_1, n_2) \qquad \text{when } H_0: t_1 = \cdots = t_v = 0 \text{ is true.}$$

Now consider

$$E\left(y_{ij} - \bar{y}_{i.}\right)^2 = E\left(e_{ij} - \bar{e}_{i.}\right)^2 = E\left(e_{ij}\right)^2 + E\left(\bar{e}_{i.}\right)^2 - 2E\left(e_{ij}\bar{e}_{i.}\right)$$
$$= \text{var}(e_{ij}) + \text{var}(\bar{e}_{i.}) - 2\text{cov}\left(e_{ij}, \bar{e}_{i.}\right) = \sigma^2 + \frac{\sigma^2}{n_i} - 2\frac{\sigma^2}{n_i}$$
$$= \sigma^2 - \frac{\sigma^2}{n_i}$$

since e_{ij} and $e_{i'j'}$ are uncorrelated for $j \neq j'$. Hence

$$E\left(y_{ij} - \bar{y}_{i.}\right)^2 = \sigma^2 - \frac{\sigma^2}{n_i}$$

Therefore

$$E\left[\sum_{i=1}^{v}\sum_{j=1}^{n_i}\left(y_{ij} - \bar{y}_{i.}\right)^2\right] = \sum_{i=1}^{v}\sum_{j=1}^{n_i}E\left(y_{ij} - \bar{y}_{i.}\right)^2 = \sum_{i=1}^{v} n_i\left(1 - \frac{1}{n_i}\right)\sigma^2 = \sum_{i=1}^{v}\left(n_i - 1\right)\sigma^2 = (n-v)\sigma^2$$

This implies that

$$W = \frac{1}{\sigma^2}\sum_{i=1}^{v}\sum_{j=1}^{n_i}\left(y_{ij} - \bar{y}_{i.}\right)^2 \sim \chi^2 \quad \text{with } n - v \text{ degrees of freedom.}$$

We note that

$$E\left\{\frac{W}{n-v}\right\} = \sigma^2,$$

i.e. the Error Sum of Squares divided by its degrees of freedom is an unbiased estimator of σ^2.

Next consider

$$\bar{y}_{i.} - \bar{y}_{..} = \left(\mu + t_i + \bar{e}_{i.}\right) - \left(\mu + \bar{t} + \bar{e}_{..}\right) = \left(t_i - \bar{t}\right) + \left(\bar{e}_{i.} - \bar{e}_{..}\right)$$

Therefore

$$\left(\bar{y}_{i.} - \bar{y}_{..}\right) \sim N\left[\left(t_i - \bar{t}\right), \text{var}\left(\bar{y}_{i.} - \bar{y}_{..}\right)\right]$$

This means that $\left(\bar{y}_{i.} - \bar{y}_{..}\right)^2$ has a non-central χ^2 with 1 degree of freedom.

When $H_0: t_1 = t_2 = \cdots = t_v = t$, say, then $\bar{t} = \frac{\sum n_i t_i}{n} = t$ and

$$\left(\bar{y}_{i.} - \bar{y}_{..}\right) \sim N\left[0, \text{var}\left(\bar{y}_{i.} - \bar{y}_{..}\right)\right] \quad \text{when } H_0: t_1 = \cdots = t_v = t \text{ is true.}$$

Now

$$E\left(\bar{y}_{i.} - \bar{y}_{..}\right)^2 = \text{var}\left(\bar{y}_{i.} - \bar{y}_{..}\right) + \left(t_i - \bar{t}\right)^2 = \left(\frac{1}{n_i} - \frac{1}{n}\right)\sigma^2 + \left(t_i - \bar{t}\right)^2$$

Hence

$$E\left[\sum_{i=1}^{v} n_i\left(\bar{y}_{i.} - \bar{y}_{..}\right)^2\right] = (v-1)\sigma^2 + \sum_{i=1}^{v} n_i\left(t_i - \bar{t}\right)^2$$

which is unbiased for $(v-1)\sigma^2$ when $H_0: t_1 = \cdots = t_v$ is true. Thus

$$U = \frac{1}{\sigma^2}\sum_{i=1}^{v} n_i\left(\bar{y}_{i.} - \bar{y}_{..}\right)^2 \sim \chi^2 \text{ with } (v-1) \text{ degrees of freedom when } H_0: t_1 = \cdots = t_v = 0 \text{ is true.}$$

To show that U and W are stochastically independent, note that $\bar{y}_{i.}$ and $\left(y_{ij} - \bar{y}_{i.}\right)$ are independent (a property of normal samples), hence U and W are independent, and so

$$F = \frac{U/(v-1)}{W/(n-v)} \sim F(n_1, n_2) \qquad \text{when } H_0: t_1 = \cdots = t_v = 0 \text{ is true,}$$

i.e. $E[\text{numerator}] = \sigma^2$ when H_0 is true and $E[\text{denominator}] = \sigma^2$ always, so that $F \to 1$ when H_0 holds and $F > 1$ otherwise.

Computational Formulas

$$TSS = \sum_{i=1}^{v}\sum_{j=1}^{n_i} y_{ij}^2 - \frac{G^2}{n}$$

where

$$G = \sum_{i=1}^{v} \sum_{j=1}^{n_i} y_{ij} \qquad \text{and} \qquad \frac{G^2}{n} \text{ is called the 'Correction Factor'.}$$

Next

$$SS\,TMT = \sum_{i=1}^{v} \frac{T_i^2}{n_i} - \frac{G^2}{n}, \qquad T_i = \sum_{j=1}^{n_i} y_{ij} = n_i \bar{y}_{i.}$$

and

$$SSE = TSS - SS\,TMT$$

Example 1

A set of data involving four 'tropical feedstuffs A, B, C, D' tried on 20 chicks is given below. All the 20 chicks are treated alike in all respects except the feeding treatment. Each feeding treatment is given to 5 chicks. Weight gains of baby chicks fed on different feeding materials composed of tropical feedstuffs are given in the table below.

Treatment	Weight gains	T_i
A	55, 49, 42, 21, 52	219
B	61, 112, 30, 89, 63	355
C	42, 97, 81, 95, 92	407
D	169, 137, 169, 85, 154	714

a. Write down the statistical model for analyzing these data. Explain all the terms in the model including the necessary assumptions on them.

b. Write down the hypothesis to be tested for comparing the effect of the feedstuffs.

c. Prepare the ANOVA table and hence complete the analysis of the data.

Solution

(a) The statistical model for analysis is given by

$$y_{ij} = \mu + t_i + e_{ij}; \qquad i = 1, 2, ..., v; \quad j = 1, 2, ..., n_i$$

where

- μ is the grand mean, i.e. the mean weight gain for all feeds if no treatments were applied,
- t_i is a constant measuring the effect of the ith treatment (feedstuff), i.e. the ith treatment either increases or decreases the weight by t_i,
- e_{ij} is the random error effect.

We shall assume further that:

- all the observations y_{ij}, $i = 1, ..., v, j = 1, ..., n_i$, are independent random variables (here all $n_i = 5$);
- the e_{ij} are independent and identically distributed as $N(0, \sigma_e^2)$, where σ_e^2 is the variance of e_{ij}.

(b) The hypothesis to be tested is

$$H_0: t_1 = t_2 = t_3 = t_4 = 0 \qquad \text{against} \qquad H_a: t_i \neq t_{i'} \text{ for some } i \neq i',$$

where $A = 1, B = 2, C = 3, D = 4$.

(c) Computations. Here $v = 4, n_i = 5$ for all $i, n = \sum n_i = 20$.

$$\sum_{i=1}^{v} \sum_{j=1}^{n_i} y_{ij}^2 = 55^2 + 49^2 + \cdots + 85^2 + 154^2 = 181,445$$

$$\text{Correction Factor} = \frac{G^2}{n} = \frac{(1695)^2}{20} = 143,651.25$$

so that

$$TSS = \sum_i \sum_j y_{ij}^2 - \frac{G^2}{n} = 181,445 - 143,651.25 = 37,793.75$$

$$SS\,TMT = \sum_{i=1}^{v} \frac{T_i^2}{n_i} - \frac{G^2}{n} = \frac{219^2}{5} + \frac{355^2}{5} + \frac{407^2}{5} + \frac{714^2}{5} - 143,651.25 = 26,234.95$$

$$\text{Error Sum of Squares } (SSE) = TSS - SS\,TMT = 37,793.75 - 26,234.95 = 11,558.80$$

ANOVA Table

Source of variation	df	SS	MSS	Variance ratio, F
Treatment	3	26,234.95	8,744.98 $= T$	$F_c = T/E = 12.105$

Source of variation	df	SS	MSS	Variance ratio, F
Error	16	11,558.80	722.42 $= E$	
Total	19	37,793.75	–	–

Now if $H_0: t_1 = t_2 = t_3 = t_4 = 0$ is true, our computed F has an F-distribution with 3 and 16 degrees of freedom respectively. At the 0.05 level of significance, tabular $F_{0.05}(3, 16) = 3.06$.

Since $F_c \geq F_{0.05}(3, 16) = 3.06$, we reject H_0 and conclude that the treatments (A, B, C and D) differ significantly.

In most cases the treatment with the greatest mean is recommended as superior. However, a t-test can be employed to decide which treatment is preferable.

If $\hat{t}_i$ and $\hat{t}_{i'}$ are the estimates of two treatment effects t_i and $t_{i'}$, then

$$\hat{t}_i - \hat{t}_{i'} = \left(\bar{y}_{i.} - \bar{y}_{..}\right) - \left(\bar{y}_{i'.} - \bar{y}_{..}\right) = \bar{y}_{i.} - \bar{y}_{i'.}$$

Now

$$E\left(\hat{t}_i - \hat{t}_{i'}\right) = 0, \qquad \operatorname{var}\left(\hat{t}_i - \hat{t}_{i'}\right) = \operatorname{var}\left(\bar{y}_{i.} - \bar{y}_{i'.}\right)$$

$$\operatorname{var}\left(\bar{y}_{i.} - \bar{y}_{i'.}\right) = \operatorname{var}\left(\bar{e}_{i.} - \bar{e}_{i'.}\right) = \left(\frac{1}{n_i} + \frac{1}{n_{i'}}\right)\sigma^2$$

If σ^2 is unknown it is estimated by the MSE, i.e.

$$\widehat{\sigma}^2 = \frac{SSE}{n - v} = MSE$$

The test statistic is the random variable

$$t = \frac{\hat{t}_i - \hat{t}_{i'}}{\sqrt{\widehat{\sigma}^2\left(\frac{1}{n_i} + \frac{1}{n_{i'}}\right)}} = \frac{\bar{y}_{i.} - \bar{y}_{i'.}}{\sqrt{\widehat{\sigma}^2\left(\frac{1}{n_i} + \frac{1}{n_{i'}}\right)}} \sim t(n - v)$$

Suppose both treatments are repeated an equal number of times, say $n_i = n_{i'} = r$; then to compare two treatments i and i' we calculate

$$t = \frac{\bar{y}_{i.} - \bar{y}_{i'.}}{\sqrt{\frac{2\widehat{\sigma}^2}{r}}} \sim t(n - v)$$

and reject $H_0: t_i = t_{i'}$ (that the two treatments are the same) at level α if $t > c$, where c is such that $P\{t > c \mid H_0 \text{ true}\} = \alpha$.

Normally we find the **critical difference (CD)**, which is the least difference between any two means:

$$CD = t_{\alpha/2,\ (\text{error d.f.})}\sqrt{\widehat{\sigma}^2\left(\frac{2}{n}\right)}$$

The treatment means are computed and compared with the CD to see whether the difference exceeds the CD (i.e. is significant).

II. CRD — Random Effects Model

For this case the treatments are a random sample drawn from a large pool of treatments and the model is given by

$$y_{ij} = \mu + t_i + e_{ij} \qquad \text{where } e_{ij} \sim N\left[0, \sigma_e^2\right], \quad t_i \sim N\left[0, \sigma_t^2\right],$$

and the e_{ij} and the t_i are mutually independent. This is equivalent to

$$y_{ij} \sim N\left[E(y_{ij}),\ \sigma_t^2 + \sigma_e^2\right]$$

e_{ij} is called the error variable.

Estimation of the components follows the same procedure as in the case of the fixed-effects model. The null hypothesis is

$$H_0: t_1 = t_2 = \cdots = t_v = 0 \qquad \Rightarrow \qquad \sigma_t^2 = 0.$$

> *[The source works through $E(SS\ TMT)$, $E(SSE)$ and $E(TSS)$ term by term across pp. 15–16; the outcome is]*

$$E(SSE) = (n - v)\sigma_e^2 \qquad \text{(always)},$$

$$E(SS\ TMT) = (v - 1)\sigma_e^2 + \left(n - \sum_i \frac{n_i^2}{n}\right)\sigma_t^2,$$

$$E(TSS) = (n - 1)\sigma_e^2 + \left(n - \sum_i \frac{n_i^2}{n}\right)\sigma_t^2.$$

So that when $H_0: t_1 = \cdots = t_v = 0$ is true, i.e. $\sigma_t^2 = 0$, both $MS\ TMT$ and $MS\ E$ estimate σ_e^2 and the ratio

$$F = \frac{MS\,TMT}{MS\,E} \sim F(v-1,\ n-v)$$

provides the test of H_0, the null hypothesis being rejected when F exceeds the tabular value.

The variance components are estimated from the mean squares by

$$\widehat{\sigma_e^2} = MS\,E, \qquad \widehat{\sigma_t^2} = \frac{MS\,TMT - MS\,E}{\left(n - \sum_i n_i^2/n\right)/(v-1)}$$

which for equal replication $n_i = n_0$ reduces to the familiar $\widehat{\sigma_t^2} = (MS\,TMT - MS\,E)/n_0$.

CHAPTER 2: RANDOMIZED BLOCK DESIGN

I. Fixed Effects Model

Introduction. In this field plan the field is first divided into blocks (constraints) such that within the blocks there is uniformity. The treatments are then assigned to each block in a random order. Such a field plan is known as a Randomized Block Design (RBD).

Block 1	t_1	t_5	⋯	t_v	t_3
Block 2	t_4	t_1	t_{v-5}	t_v	t_2
Block 3	t_v	t_6	t_3	t_5	t_2

Figure. A layout of RBD.

The above figure depicts a field plan of a Randomized Block Design with v treatments and k blocks. In this plan no treatment appears more than once in any one block. Each treatment occurs with probability $\frac{1}{v}$ in each block.

In an industrial setting, blocks can be thought of as shifts, e.g. the morning shift, etc. The blocks can also be looked at as the conditions under which the workers are working, e.g. temperature (some workers are provided with air conditioning while others are not), etc.

Let there be v treatments and k blocks. Let y_{ij} denote the observation of the ith treatment in the jth block ($i = 1, 2, \ldots, v$; $j = 1, 2, \ldots, k$). Assume

$$y_{ij} = \mu + t_i + b_j + e_{ij}$$

$$\text{var}(y_{ij}) = \sigma^2 \text{ a constant}, \qquad \text{cov}(y_{ij}, y_{i'j'}) = 0, \quad (i,j) \neq (i',j')$$

where

- μ is the grand mean, i.e. the mean yield if no treatments and no blocking were applied,
- t_i is a constant measuring the effect of the ith treatment,
- b_j is a constant measuring the effect of the jth block,
- e_{ij} is the random error effect.

The model of analysis is known as a **Linear Fixed Effect Additive Model**.

The hypotheses of interest in this case are:

i. $H_0: t_1 = t_2 = \cdots = t_v = 0$ against $H_a: t_i \neq t_{i'}$ for some $i \neq i'$

ii. $H_0: b_1 = b_2 = \cdots = b_k = 0$ against $H_a: b_j \neq b_{j'}$ for some $j \neq j'$

Analysis Procedure

Step I: Estimation of effects μ, t_i, b_j such that

$$S = \sum_{i=1}^{v} \sum_{j=1}^{k} \left[y_{ij} - \left(\mu + t_i + b_j \right) \right]^2$$

is minimized. The conditions of minimum are:

1. $\frac{\partial S}{\partial \mu} = 0$
2. $\frac{\partial S}{\partial t_i} = 0, i = 1, 2, \ldots, v$
3. $\frac{\partial S}{\partial b_j} = 0, j = 1, 2, \ldots, k$

Equations 1)–3) give rise to $(1 + v + k)$ equations:

$$1) \text{ gives } \sum_{i=1}^{v} \sum_{j=1}^{k} \left(y_{ij} - \mu - t_i - b_j \right) = 0$$

$$2) \text{ gives } \sum_{j=1}^{k} \left(y_{ij} - \mu - t_i - b_j \right) = 0$$

$$3) \text{ gives } \sum_{i=1}^{v} \left(y_{ij} - \mu - t_i - b_j \right) = 0$$

Summing equation 2) over i we get equation 1). Similarly, summing equation 3) over j we get equation 1). So the two equations are linearly dependent on equation 1). Thus the number of independent equations is $(1 + v + k) - 2$, which means that the solutions to these equations are not unique.

For uniqueness of the solutions we add two independent equations, namely

$$\sum_{i=1}^{v} t_i = 0, \qquad \sum_{j=1}^{k} b_j = 0.$$

Note that the solutions will depend on the choice of these additional equations, but the analysis remains invariant.

The solutions are:

1. $\sum_{i=1}^{v}\sum_{j=1}^{k} y_{ij} = \sum_i \sum_j \mu + \sum_i \sum_j t_i + \sum_i \sum_j b_j$, which gives

$$\hat{\mu} = \bar{y}_{..} = \frac{G}{vk} = \frac{1}{vk}\sum_{i=1}^{v}\sum_{j=1}^{k} y_{ij}$$

2. $\sum_{i=1}^{v} y_{ij} = k\mu + k\,t_i + \sum_i b_j = k\mu + k\,t_i$ gives $\hat{t}_i = \bar{y}_{i.} - \bar{y}_{..}$
3. $\sum_{j=1}^{k} y_{ij} = v\mu + v\,b_j$ gives $\hat{b}_j = \bar{y}_{.j} - \bar{y}_{..}$, where $\bar{y}_{.j} = \frac{1}{v}\sum_i y_{ij}$

Plugging these values in the model we get

$$y_{ij} = \bar{y}_{..} + \left(\bar{y}_{i.} - \bar{y}_{..}\right) + \left(\bar{y}_{.j} - \bar{y}_{..}\right) + \left(y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..}\right)$$

The model can then be written as

$$y_{ij} = \mu + t_i + b_j + e_{ij}$$

where

$$E(e_{ij}) = 0, \qquad \text{var}(e_{ij}) = \sigma^2 \text{ a constant}, \qquad \text{cov}(e_{ij}, e_{i'j'}) = 0, \quad (i,j) \neq (i',j').$$

For the purpose of analysis assume further that

$$e_{ij} \sim N(0,\sigma^2) \qquad \Leftrightarrow \qquad y_{ij} \sim N\left(\mu + t_i + b_j,\ \sigma^2\right)$$

Let

$$y_{ij} - \bar{y}_{..} = \left(\bar{y}_{i.} - \bar{y}_{..}\right) + \left(\bar{y}_{.j} - \bar{y}_{..}\right) + \left(y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..}\right)$$

Squaring and summing both sides we get

$$\sum_i \sum_j \left(y_{ij} - \bar{y}_{..}\right)^2 = \sum_i \sum_j \left(\bar{y}_{i.} - \bar{y}_{..}\right)^2 + \sum_i \sum_j \left(\bar{y}_{.j} - \bar{y}_{..}\right)^2 + \sum_i \sum_j \left(y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..}\right)^2$$

(cross-product terms vanish). Thus

$$(TSS) = (SS\,TMT) + (SS\,BLOCK) + (SSE)$$

We can therefore form the ANOVA table as follows.

ANOVA table

Source of variation	df	SS	MSS	F
Treatment	$v-1$	$k\sum_{i=1}^{v}\left(\bar{y}_{i.} - \bar{y}_{..}\right)^2$	$T = \frac{SS\,TMT}{v-1}$	$F_1 = \frac{T}{E}$
Block	$k-1$	$v\sum_{j=1}^{k}\left(\bar{y}_{.j} - \bar{y}_{..}\right)^2$	$B = \frac{SS\,BLOCK}{k-1}$	$F_2 = \frac{B}{E}$
Error	$(v-1)(k-1)$	$\sum_i \sum_j \left(y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..}\right)^2$	$E = \frac{SSE}{(v-1)(k-1)}$	
Total	$vk-1$	$\sum_{i=1}^{v}\sum_{j=1}^{k}\left(y_{ij} - \bar{y}_{..}\right)^2$	–	

To test

i. $H_0: t_1 = t_2 = \cdots = t_v = 0$ against $H_a: t_i \neq t_{i'}$ for some $i \neq i'$,

calculate $F_1 = T/E$ and reject H_0 at level α if $F_1 > F_\alpha(n_1, n_2)$ tabular value, where

$$n_1 = v-1, \qquad n_2 = (v-1)(k-1);$$

ii. $H_0: b_1 = b_2 = \cdots = b_k = 0$ against $H_a: b_j \neq b_{j'}$ for some $j \neq j'$,

calculate $F_2 = B/E$ and reject H_0 at level α if $F_2 > F_\alpha(n_1, n_2)$ tabular value, where

$$n_1 = k-1, \qquad n_2 = (v-1)(k-1).$$

Computation formulas

Total sum of squares:

$$TSS = \sum_{i=1}^{v}\sum_{j=1}^{k}\left(y_{ij} - \bar{y}_{..}\right)^2 = \sum_{i=1}^{v}\sum_{j=1}^{k} y_{ij}^2 - \frac{G^2}{vk}, \qquad \text{where } G = \sum_{i=1}^{v}\sum_{j=1}^{k} y_{ij}$$

Treatment sum of squares:

$$SS\,TMT = k\sum_{i=1}^{v}\left(\bar{y}_{i.}-\bar{y}_{..}\right)^2 = \sum_{i=1}^{v}\frac{T_i^2}{k}-\frac{G^2}{vk}, \qquad \text{where } T_i=\sum_{j=1}^{k} y_{ij}$$

Block sum of squares:

$$SS\,BLOCK = v\sum_{j=1}^{k}\left(\bar{y}_{.j}-\bar{y}_{..}\right)^2 = \sum_{j=1}^{k}\frac{B_j^2}{v}-\frac{G^2}{vk}, \qquad \text{where } B_j=\sum_{i=1}^{v} y_{ij}$$

Error sum of squares is obtained by subtraction, i.e.

$$SSE = TSS-(SS\,TMT)-(SS\,BLOCK)$$

Example 2

Three kinds of gasoline were used in four makes of automobiles. Do the values (miles/gallon) indicate a significant difference among the kinds of gasoline or among the makes of automobiles?

	M1	M2	M3	M4	T_i
A1	14.6	15.2	15.1	14.6	$T_1 = 59.5$
A2	15.3	14.8	17.0	16.1	$T_2 = 63.2$
A3	17.2	16.9	17.1	16.6	$T_3 = 68.2$
	$B_1 = 47.1$	$B_2 = 46.9$	$B_3 = 49.6$	$B_4 = 47.3$	$G = 190.9$

Solution

Identify gasoline with treatments (A_1, A_2, A_3) and automobiles with blocks (M_1, M_2, M_3, M_4). The model of analysis is

$$y_{ij} = \mu + t_i + b_j + e_{ij}, \qquad i = 1,2,3; \quad j = 1,2,3,4$$

where

- μ is the grand mean, i.e. the mean mileage/gallon for all gasoline and all automobiles,
- t_i is a constant measuring the effect of the ith treatment (gasoline),
- b_j is a constant measuring the effect of the jth block (automobile),
- e_{ij} is the random error effect,

with $E(e_{ij}) = 0$, $\text{var}(e_{ij}) = \sigma^2$ a constant, $\text{cov}(e_{ij}, e_{i'j'}) = 0$ for $(i,j) \neq (i', j')$. For the purpose of analysis assume further that

$$e_{ij} \sim N(0,\sigma^2) \qquad \Leftrightarrow \qquad y_{ij} \sim N\left(\mu + t_i + b_j,\ \sigma^2\right)$$

We wish to test

i. $H_0: t_1 = t_2 = t_3$ against $H_a: t_i \neq t_{i'}$ for some $i \neq i'$

ii. $H_0: b_1 = b_2 = b_3 = b_4$ against $H_a: b_j \neq b_{j'}$ for some $j \neq j'$

Computations:

$$TSS = \sum_{i=1}^{3}\sum_{j=1}^{4} y_{ij}^2 - \frac{G^2}{12} = (14.6)^2 + (15.2)^2 + \cdots + (17.1)^2 + (16.6)^2 - \frac{(190.9)^2}{12}$$

$$= 3049.97 - 3036.900 = 13.07$$

$$SS\,TMT = \frac{1}{4}\left(T_1^2 + T_2^2 + T_3^2\right) - \frac{G^2}{12} = \frac{(59.5)^2 + (63.2)^2 + (68.2)^2}{4} - \frac{(190.9)^2}{12}$$

$$= 3046.435 - 3036.900 = 9.535$$

$$SS\,BLOCK = \frac{1}{3}\left(B_1^2 + B_2^2 + B_3^2 + B_4^2\right) - \frac{G^2}{12} = \frac{(47.1)^2 + (46.9)^2 + (49.6)^2 + (47.3)^2}{3} - \frac{(190.9)^2}{12}$$

$$= 3038.49 - 3036.900 = 1.59$$

$$SSE = TSS - (SS\,TMT) - (SS\,BLOCK) = 13.07 - 9.535 - 1.59 = 1.945$$

We then form the ANOVA table.

ANOVA table

Source of variation	df	SS	MSS	F
Treatment	2	9.535	4.768	$F = 4.768/0.324 = 14.72$
Block	3	1.59	0.53	$F = 0.53/0.324 = 1.64$
Error	6	1.945	0.324	
Total	11	13.07	–	

i. To test $H_0: t_1 = t_2 = t_3$ against $H_a: t_i \neq t_{i'}$ for some $i \neq i'$: we see that $F_1 = 14.72 > F_{0.05}(2,6) = 5.14$, so we reject H_0 and conclude that the gasoline kinds are different.

ii. To test $H_0: b_1 = b_2 = b_3 = b_4$ against $H_a: b_j \neq b_{j'}$ for some $j \neq j'$: we see that $F_2 = 1.64 < F_{0.05}(3,6) = 4.76$, so we fail to reject H_0 and conclude that the automobiles are not different.

II. RBD — Random Effects Model

In this case the treatments and blocks are samples from larger populations of treatments and blocks respectively. The model is given by

$$y_{ij} = \mu + t_i + b_j + e_{ij}, \qquad i = 1, 2, ..., v, \quad j = 1, 2, ..., b$$

where μ is the grand mean,

$$t_i \sim N\left(0, \sigma_t^2\right), \quad \text{cov}(t_i, t_{i'}) = 0; \qquad b_j \sim N\left(0, \sigma_b^2\right), \quad \text{cov}(b_j, b_{j'}) = 0;$$
$$e_{ij} \sim N\left(0, \sigma_e^2\right), \quad \text{cov}\left(e_{ij}, e_{i'j'}\right) = 0.$$

Treatments, blocks and error term are all mutually independent.

To obtain the estimates of the parameters we seek the values that minimize

$$S = \sum_{i=1}^{v} \sum_{j=1}^{b} \left[y_{ij} - \left(\mu + t_i + b_j\right)\right]^2.$$

The least-squares estimates are given as in the case of the fixed-effects model above, with all the assumptions such as uniqueness of the solutions, where we add two independent equations, namely

$$\sum_{i=1}^{v} t_i = 0, \qquad \sum_{j=1}^{b} b_j = 0.$$

> *[The source evaluates the expectations of the sums of squares term by term over pp. 23–25. The results are]*

$$E\left(SS\,TMT\right) = (v-1)\sigma_e^2 + (bv - v)\sigma_t^2 = (v-1)\left(\sigma_e^2 + b\,\sigma_t^2\right),$$
$$E\left(SS\,BLOCK\right) = (b-1)\sigma_e^2 + (vb - b)\sigma_b^2 = (b-1)\left(\sigma_e^2 + v\,\sigma_b^2\right),$$
$$E\left(SSE\right) = (v-1)(b-1)\sigma_e^2,$$
$$E\left(TSS\right) = \sigma_e^2(vb-1) + b(v-1)\sigma_t^2 + v(b-1)\sigma_b^2.$$

Therefore under

$$H_0: t_1 = t_2 = \cdots = t_v = 0 \Rightarrow \sigma_t^2 = 0, \qquad H_0: b_1 = b_2 = \cdots = b_b = 0 \Rightarrow \sigma_b^2 = 0,$$

we have

$$E\left\{\frac{SS\,TMT}{v-1}\right\} = \sigma_e^2 \qquad \text{and} \qquad E\left\{\frac{SS\,BLOCK}{b-1}\right\} = \sigma_e^2 \qquad \text{respectively,}$$

so that the tests for treatment and block equality are based respectively on the statistics

$$F_1 = \frac{SS\,TMT/(v-1)}{SSE/[(v-1)(b-1)]} \sim F\left[(v-1),\ (v-1)(b-1)\right] \quad \text{under } H_0: t_1 = \cdots = t_v = 0,$$

and

$$F_2 = \frac{SS\,BLOCK/(b-1)}{SSE/[(v-1)(b-1)]} \sim F\left[(b-1),\ (v-1)(b-1)\right] \quad \text{under } H_0: b_1 = \cdots = b_b = 0,$$

and the tests are to reject H_0 whenever $F_1 > F_{c_1}$ and $F_2 > F_{c_2}$ respectively.

Again, as in the case of the fixed-effects model, we can form the ANOVA table as follows.

ANOVA table

Source of variation	df	SS	MSS	F
Treatment	$v-1$	$k\sum_i \left(\bar{y}_{i.} - \bar{y}_{..}\right)^2$	T	$F_1 = \frac{T}{E}$
Block	$b-1$	$v\sum_j \left(\bar{y}_{.j} - \bar{y}_{..}\right)^2$	B	$F_2 = \frac{B}{E}$
Error	$(v-1)(b-1)$	$\sum_i \sum_j \left(y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..}\right)^2$	E	
Total	$vb-1$	$\sum_i \sum_j \left(y_{ij} - \bar{y}_{..}\right)^2$	-	

Example 3

The following data are the results from a randomized block design experiment in which the treatments were drawn from some large pool of treatments and blocks too were drawn from a large pool of blocks.

	Treatment 1	Treatment 2	Treatment 3	Treatment 4	
Block 1	6	8	7	11	32
Block 2	5	6	6	8	25
Block 3	9	9	11	12	41
	20	23	24	31	98

Test the null hypotheses of no treatment and block effects (use $\alpha = 0.01$).

Solution

The null hypotheses are

$$H_0: t_1 = t_2 = t_3 = t_4 \qquad \text{and} \qquad H_0: b_1 = b_2 = b_3.$$

$$TSS = \sum_{i=1}^{4}\sum_{j=1}^{3} y_{ij}^2 - \frac{G^2}{12} = \left(6^2 + 8^2 + \cdots + 11^2 + 12^2\right) - \frac{98^2}{12} = 57.7$$

$$SS\,TMT = \sum_{i=1}^{4} \frac{T_i^2}{3} - \frac{G^2}{12} = \frac{20^2 + 23^2 + 24^2 + 31^2}{3} - \frac{98^2}{12} = 21.7$$

$$SS\,BLOCK = \sum_{j=1}^{3} \frac{B_j^2}{4} - \frac{G^2}{12} = \frac{32^2 + 25^2 + 41^2}{4} - \frac{98^2}{12} = 32.2$$

$$SSE = TSS - (SS\,TMT) - (SS\,BLOCK) = 57.7 - 21.7 - 32.2 = 3.8$$

ANOVA table

Source of variation	df	SS	MSS	F
Treatment	3	21.7	7.23	$F = 7.23/0.63 = 11.48$
Block	2	32.2	16.1	$F = 16.1/0.63 = 25.56$
Error	6	3.8	0.63	
Total	11	57.7	–	

Now

$$F_{0.01}[3, 6] = 9.78, \qquad F_{0.01}[2, 6] = 10.92.$$

Since $F_1 > F_{0.01}[3, 6]$, reject H_0 and conclude that the treatment effects are significantly different at the $\alpha = 0.01$ level. Also, since $F_2 > F_{0.01}[2, 6]$, reject H_0 and conclude that the block effects are significantly different at the 0.01 level.

CHAPTER 3: ESTIMATION OF A MISSING OBSERVATION IN A RBD

Let the ith treatment in the jth block be missing, i.e. y_{ij} is missing:

	B_1	B_2	$\cdots$	B_j	$\cdots$	B_b
T_1	y_{11}	y_{12}	$\cdots$	y_{1j}	$\cdots$	y_{1b}
T_2	y_{21}	y_{22}	$\cdots$	y_{2j}	$\cdots$	y_{2b}
$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$
T_i	y_{i1}	y_{i2}	$\cdots$	y_{ij}	$\cdots$	y_{ib}
$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$
T_v	y_{v1}	y_{v2}	$\cdots$	y_{vj}	$\cdots$	y_{vb}

Let us denote the missing observation by X. The problem is to estimate X.

$$TSS = \sum_{i=1}^{v}\sum_{j=1}^{b} y_{ij}^2 - \frac{G^2}{vb} = \sum_{i\neq l}^{v}\sum_{j\neq k}^{b} y_{ij}^2 + X^2 - \frac{(G^{'}+X)^2}{vb}$$

$$= X^2 - \frac{2G^{'}X}{vb} + \text{terms independent of } X$$

where $G^{'}$ is the total of all known observations.

$$SS\,TMT = b\sum_{i=1}^{v}\left(\bar{y}_{i.} - \bar{y}_{..}\right)^2 = \sum_{i}\frac{T_i^2}{b} - \frac{G^2}{vb} = \frac{(T_i^{'}+X)^2}{b} + \sum_{i\neq k}\frac{T_i^{'2}}{b} - \frac{(G^{'}+X)^2}{vb}$$

$$= \frac{(T_i^{'}+X)^2}{b} - \frac{(G^{'}+X)^2}{vb} + \text{terms independent of } X$$

where $T_i^{'}$ is the total of all known observations receiving the ith treatment.

$$SS\,BLOCK = v\sum_{j=1}^{b}\left(\bar{y}_{.j} - \bar{y}_{..}\right)^2 = \sum_{j}\frac{B_j^2}{v} - \frac{G^2}{vb} = \frac{(B_j^{'}+X)^2}{v} - \frac{(G^{'}+X)^2}{vb} + \text{terms independent of } X$$

where $B_j^{'}$ is the total of all known observations in the jth block.

$$SSE = TSS - (SS\,TMT) - (SS\,BLOCK)$$

$$= X^2 - \frac{2X(G^{'}+X)}{vb} - \frac{(T_i^{'}+X)^2}{b} - \frac{(B_j^{'}+X)^2}{v} + \text{terms independent of } X$$

We determine X which minimizes SSE. Differentiating SSE with respect to X and equating to zero gives

$$2X - \frac{2(G^{'}+X)}{vb} - \frac{2(T_i^{'}+X)}{b} - \frac{2(B_j^{'}+X)}{v} = 0$$

$$X\left(1 - \frac{1}{vb} - \frac{1}{b} - \frac{1}{v}\right) - \frac{G^{'}}{vb} - \frac{T_i^{'}}{b} - \frac{B_j^{'}}{v} = 0$$

$$X\,(\,vb - 1 - v - b\,) = G^{'} + vT_i^{'} + bB_j^{'}$$

That is

$$X = \frac{vT_i^{'} + bB_j^{'} - G^{'}}{(v-1)(b-1)}$$

Note: when the missing observation is used to calculate $SS\,TMT$ then it can be shown that $SS\,TMT$ is overestimated by an amount

$$Bias = \frac{\left(B_j^{'} + vT_i^{'} - G^{'}\right)^2}{v^2(v-1)(b-1)}$$

Example

The following is data for a RBD with a missing observation for treatment 1 in block 2. Carry out the analysis after estimating the missing observation (use $\alpha = 0.05$).

Treatment	B1	B2	B3	B4
T1	19	–	23	26
T2	26	28	27	33
T3	20	29	22	26

Solution

We wish to test

$$H_0: t_1 = t_2 = t_3 \qquad \text{and} \qquad H_0: b_1 = b_2 = b_3 = b_4.$$

Treatment totals: $T_1^{'} = 68, T_2 = 114, T_3 = 97.$

Block totals: $B_1^{'} = 65, B_2^{'} = 57, B_3^{'} = 72, B_4^{'} = 85, G^{'} = 279.$

The missing observation is estimated by

$$X = \frac{vT_i^{'} + bB_j^{'} - G^{'}}{(v-1)(b-1)} = \frac{3\times 68 + 4\times 57 - 279}{(4-1)(3-1)} = \frac{204 + 228 - 279}{6} = 25.5$$

Thus

$$T_1 = 68 + 25.5 = 93.5, \qquad B_2 = 57 + 25.5 = 82.5, \qquad G = 279 + 25.5 = 304.5$$

$$\frac{G^2}{vb} = \frac{(304.5)^2}{12} = 7726.7$$

$$TSS = \sum_i \sum_j y_{ij}^2 - \frac{G^2}{vb} = 7895.25 - 7726.7 = 168.55$$

$$SS\,TMT = \sum_{i=1}^{3} \frac{T_i^2}{4} - \frac{G^2}{12} = 7786.8 - 7726.7 = 60.1$$

$$SS\,BLOCK = \sum_{j=1}^{4} \frac{B_j^2}{3} - \frac{G^2}{12} = 7813.4 - 7726.7 = 86.7$$

$$SSE = TSS - (SS\,TMT) - (SS\,BLOCK) = 168.55 - 60.1 - 86.7 = 21.75$$

$$Bias = \frac{\left(B_j^{'} + vT_i^{'} - G^{'}\right)^2}{v^2(v-1)(b-1)} = \frac{(57 + 3\times 68 - 279)^2}{3^2(3-1)(4-1)} = \frac{(-18)^2}{54} = 6$$

So the adjusted treatment sum of squares is

$$SS\,TMT_{adj} = 60.1 - 6 = 54.1$$

ANOVA Table

Source of variation	d.f.	SS	MSS	F-ratio
Treatment (adjusted)	2	54.1	27.05	$F = 27.05/4.35 = 6.21$
Blocks	3	86.7	28.9	$F = 28.9/4.35 = 6.64$
Error	5	21.75	4.35	
Total	10	168.55		

We have $F_1 = 6.21 > F_{0.05}[2,5] = 5.79$, hence we reject H_0 at the 0.05 level and conclude that the treatments are not the same; and $F_2 = 6.64 > F_{0.05}[3,5] = 5.41$, hence we reject H_0 at the 0.05 level and conclude that the blocks are not the same.

Note: one degree of freedom is lost from TSS because of the estimation of one missing observation.

Exercise

The following table gives results from a RBD with 4 blocks, 3 treatments and one missing observation.

Blocks	1	2	3
1	18	27	22
2	20	15	21
3	15	21	–
4	32	20	25

Estimate the missing observation and analyze the data (use $\alpha = 0.05$).

Note: if the treatments show a significant difference in their effect then the standard error (S.E.) of the difference is:

i. $\sqrt{\frac{2\hat{\sigma}^2}{b}}$ if none of the treatments contains the missing value;

ii. $\sqrt{\hat{\sigma}^2\left(\frac{2}{b} + \frac{2}{b(b-1)(v-1)}\right)}$ if one of the treatments corresponds to the missing observation.

CHAPTER 4: LATIN SQUARE DESIGN

In this experiment the field is a $(v \times v)$ square. The 'v' treatments are assigned to the v^2 plots in 'v' rows and 'v' columns such that each treatment occurs once and only once in each row and each column.

	C_1	C_2	$\cdots$	C_v
R_1	1	2	$\cdots$	v
R_2	$v+1$	$v+2$	$\cdots$	$2v$
$\vdots$	$\vdots$	$\vdots$		$\vdots$
R_v	$2v+1$	$2v+2$	$\cdots$	v^2

Figure 4.1. A $v \times v$ Latin square.

A	B	C
B	C	A
C	A	B

Figure 4.2. A 3×3 Latin square.

A	B	C	D
B	C	D	A
C	D	A	B
D	A	B	C

Figure 4.3. A 4×4 Latin square.

The model of analysis for a $v \times v$ Latin square design is as follows. Let $y_{(ijk)}$ be the observation of treatment 'k' occurring in the ith row and the jth column $(i = 1, 2, \ldots, v;\ j = 1, 2, \ldots, v;\ k = 1, 2, \ldots, v)$. The model of analysis is

$$E\left(y_{(ijk)}\right) = \mu + r_i + c_j + t_k$$

$$\text{var}\left(y_{(ijk)}\right) = \sigma^2 \text{ a constant}, \qquad \text{cov}\left(y_{(ijk)}, y_{(i'j'k')}\right) = 0, \quad (i,j,k) \neq (i',j',k')$$

where

- μ is some unknown constant (usually the grand mean),
- r_i is a constant measuring the effect of the ith row,
- c_j is a constant measuring the effect of the jth column,
- t_k is a constant measuring the effect of the kth treatment,
- $e_{(ijk)}$ is the random error, assumed to be normally distributed with mean zero and variance σ^2.

The model of analysis is a **linear fixed-effect additive model**.

We wish to test

i. $H_0: r_1 = r_2 = \cdots = r_v$ against $H_a: r_i \neq r_{i'}$ for some $i \neq i'$
ii. $H_0: c_1 = c_2 = \cdots = c_v$ against $H_a: c_j \neq c_{j'}$ for some $j \neq j'$
iii. $H_0: t_1 = t_2 = \cdots = t_v$ against $H_a: t_k \neq t_{k'}$ for some $k \neq k'$

Analysis procedure

Step I: Estimation of parameters. The method of least squares requires a choice of μ, r_i, c_j, t_k such that

$$S = \sum_{i=1}^{v} \sum_{j=1}^{v} \left[y_{(ijk)} - \left(\mu + r_i + c_j + t_k\right)\right]^2$$

is minimized. The conditions of minimum are:

1. $\frac{\partial S}{\partial \mu} = 0$
2. $\frac{\partial S}{\partial r_i} = 0, i = 1, 2, \ldots, v$
3. $\frac{\partial S}{\partial c_j} = 0, j = 1, 2, \ldots, v$
4. $\frac{\partial S}{\partial t_k} = 0, k = 1, 2, \ldots, v$

Now

$$1) \text{ gives } \sum_{i=1}^{v}\sum_{j=1}^{v}\left(y_{(ijk)}-\mu-r_i-c_j-t_k\right)=0$$

$$2) \text{ gives } \sum_{j=1}^{v}\left(y_{(ijk)}-\mu-r_i-c_j-t_k\right)=0$$

$$3) \text{ gives } \sum_{i=1}^{v}\left(y_{(ijk)}-\mu-r_i-c_j-t_k\right)=0$$

$$4) \sum_{\text{plots holding } t_k}\left(y_{(ijk)}-\mu-r_i-c_j-t_k\right)=0$$

The number of equations/unknown constants is $(3v+1)$. Not all the equations are independent; the number of independent equations is $(3v+1)-3$.

For uniqueness of solutions we take three additional independent equations, namely

$$\sum_{i=1}^{v} r_i = 0, \qquad \sum_{j=1}^{v} c_j = 0, \qquad \sum_{\text{plots holding } t_k} t_k = 0$$

Subject to these conditions,

$$1) \quad \Rightarrow \quad \hat{\mu} = \bar{y}_{...} = \frac{G}{v^2} = \frac{1}{v^2}\sum_{i=1}^{v}\sum_{j=1}^{v} y_{(ijk)}$$

$$2) \quad \Rightarrow \quad \hat{r}_i = \bar{y}_{i..} - \bar{y}_{...}$$

$$3) \quad \Rightarrow \quad \hat{c}_j = \bar{y}_{.j.} - \bar{y}_{...}$$

$$4) \quad \Rightarrow \quad \hat{t}_k = \bar{y}_{..k} - \bar{y}_{...}, \qquad \text{where } \bar{y}_{..k} = \frac{T_k}{v}, \text{ the average for plots occupied by the } k^{\text{th}} \text{ treatment.}$$

Plugging these into the model we have

$$y_{(ijk)} = \bar{y}_{...} + \left(\bar{y}_{i..} - \bar{y}_{...}\right) + \left(\bar{y}_{.j.} - \bar{y}_{...}\right) + \left(\bar{y}_{..k} - \bar{y}_{...}\right) + \left(y_{(ijk)} - \bar{y}_{i..} - \bar{y}_{.j.} - \bar{y}_{..k} + 2\bar{y}_{...}\right)$$

This can be written as

$$y_{(ijk)} - \bar{y}_{...} = \left(\bar{y}_{i..} - \bar{y}_{...}\right) + \left(\bar{y}_{.j.} - \bar{y}_{...}\right) + \left(\bar{y}_{..k} - \bar{y}_{...}\right) + \left(y_{(ijk)} - \bar{y}_{i..} - \bar{y}_{.j.} - \bar{y}_{..k} + 2\bar{y}_{...}\right)$$

Squaring both sides and summing we get

$$\sum_i\sum_j\left(y_{(ijk)}-\bar{y}_{...}\right)^2 = \sum_i\sum_j\left(\bar{y}_{i..}-\bar{y}_{...}\right)^2 + \sum_i\sum_j\left(\bar{y}_{.j.}-\bar{y}_{...}\right)^2 + \sum_i\sum_j\left(\bar{y}_{..k}-\bar{y}_{...}\right)^2 + \sum_i\sum_j\left(y_{(ijk)}-\bar{y}_{i..}-\bar{y}_{.j.}-\bar{y}_{..k}+2\bar{y}_{...}\right)^2$$

(cross-product terms vanish). Thus

$$(TSS) = (SS\ Rows) + (SS\ Columns) + (SS\ TMT) + (SSE)$$

We can then form the ANOVA table.

Source	df	SS	MSS	F-ratio
Rows	$v-1$	$\sum_i\sum_j\left(\bar{y}_{i..}-\bar{y}_{...}\right)^2$	$R=\frac{SS\ Rows}{v-1}$	$F_r=\frac{R}{E}$
Columns	$v-1$	$\sum_i\sum_j\left(\bar{y}_{.j.}-\bar{y}_{...}\right)^2$	$C=\frac{SS\ Columns}{v-1}$	$F_c=\frac{C}{E}$
Treatment	$v-1$	$\sum_i\sum_j\left(\bar{y}_{..k}-\bar{y}_{...}\right)^2$	$T=\frac{SS\ TMT}{v-1}$	$F_t=\frac{T}{E}$
Error	$(v-1)(v-2)$	$\sum_i\sum_j\left(y_{(ijk)}-\bar{y}_{i..}-\bar{y}_{.j.}-\bar{y}_{..k}+2\bar{y}_{...}\right)^2$	$E=\frac{SSE}{(v-1)(v-2)}$	
Total	v^2-1	$\sum_i\sum_j\left(y_{(ijk)}-\bar{y}_{...}\right)^2$	–	

i. To test $H_0: t_1 = t_2 = \cdots = t_v$ against $H_a: t_k \neq t_{k'}$ for some $k \neq k'$: compute $F_t = T/E$ and reject H_0 at level α if $F_t > F_\alpha[\,v-1,\ (v-1)(v-2)\,]$ tabular value.

ii. To test $H_0: r_1 = r_2 = \cdots = r_v$ against $H_a: r_i \neq r_{i'}$ for some $i \neq i'$: compute $F_r = R/E$ and reject H_0 at level α if $F_r > F_\alpha[\,v-1,\ (v-1)(v-2)\,]$ tabular value.

iii. To test $H_0: c_1 = c_2 = \cdots = c_v$ against $H_a: c_j \neq c_{j'}$ for some $j \neq j'$: compute $F_c = C/E$ and reject H_0 at level α if $F_c > F_\alpha[\,v-1,\ (v-1)(v-2)\,]$ tabular value.

Next, using the model assumptions, one shows that

I. $U = \sum_i\sum_j\left(\bar{y}_{..k}-\bar{y}_{...}\right)^2 \sim \sigma^2\chi^2$ with $(v-1)$ degrees of freedom when $H_0: t_1 = \cdots = t_v = 0$ is true;

II. $W = \sum_i\sum_j\left(y_{(ijk)}-\bar{y}_{i..}-\bar{y}_{.j.}-\bar{y}_{..k}+2\bar{y}_{...}\right)^2 \sim \sigma^2\chi^2$ with $(v-1)(v-2)$ degrees of freedom;

III. U and W are stochastically independent.

Hence

$$F = \frac{U/(v-1)}{W/[(v-1)(v-2)]} \sim F[(v-1),\ (v-1)(v-2)] \qquad \text{when } H_0: t_1 = \cdots = t_v \text{ is true.}$$

From the properties of the χ^2 distribution one needs to show that

- $E(U) = \sigma^2(v-1)$ when $H_0: t_1 = \cdots = t_v = 0$ is true,
- $E(W) = \sigma^2(v-1)(v-2)$ always,
- $E\left(\frac{U}{v-1}\right) = \sigma^2$ when $H_0: t_1 = \cdots = t_v = 0$ is true,
- $E\left(\frac{W}{(v-1)(v-2)}\right) = \sigma^2$ always,
- $F = 1$ when $H_0: t_1 = \cdots = t_v = 0$ is true and $F > 1$ when H_0 is false.

Exercise. Show that

a. when $H_0: r_1 = \cdots = r_v$ is true then $E(R) = \sigma^2$;

b. when $H_0: c_1 = \cdots = c_v$ is true then $E(C) = \sigma^2$;

c. when $H_0: t_1 = \cdots = t_v$ is true then $E(T) = \sigma^2$;

d. $E(E) = \sigma^2$ always.

Computational formulas

1. $TSS = \sum_{i=1}^{v} \sum_{j=1}^{v} \left(y_{(ijk)}\right)^2 - \frac{G^2}{v^2}$
2. $SS\ ROW = \sum_{i=1}^{v} \frac{R_i^2}{v} - \frac{G^2}{v^2}$
3. $SS\ COL = \sum_{j=1}^{v} \frac{C_j^2}{v} - \frac{G^2}{v^2}$
4. $SS\ TMT = \sum_{k=1}^{v} \frac{T_k^2}{v} - \frac{G^2}{v^2}$

where

- R_i = total of observations in the ith row,
- C_j = total of observations in the jth column,
- T_k = total of observations for the kth treatment,
- G = total of all observations.

Remarks

1. If we regard the rows of an LSD as the blocks, then an LSD will be more efficient than a corresponding RBD if the columns are significantly different, i.e. if $C/E >$ (tabular).
2. If we regard the columns of an LSD as the blocks, then an LSD will be more efficient than a corresponding RBD if the rows are significantly different, i.e. if $R/E >$ (tabular).
3. A Latin square design will perform better than a CRD if either the rows or the columns or both are significantly different.

Example 3

The following table gives the observations from a 4×4 Latin square experiment, where A1 = treatment 1, A2 = treatment 2, A3 = treatment 3, A4 = treatment 4.

R_i					
	A1(10)	A2(8)	A3(5)	A4(4)	27
	A2(11)	A4(13)	A1(16)	A3(12)	52
	A3(10)	A1(14)	A4(9)	A2(10)	43
	A4(8)	A3(6)	A2(11)	A1(13)	38
C	39	41	41	39	160

Carry out the ANOVA to test the effects of the columns, the rows and the treatments.

Solution

Treatment totals are: $T_1 = 53, T_2 = 42, T_3 = 33, T_4 = 34$.

Source	df	SS	MSS	F-ratio
Treatment	3	63.5	21.77	$F_t = 7.94$
Row	3	81.5	27.17	$F_r = 10.9$
Column	3	1.0	0.333	$F_c = 0.12$
Error	6	16	2.67	
Total	15	162		

From the F-tables we have $F_{0.05}(3, 6) = 4.76$.

1. It is clear that we reject $H_0: t_1 = t_2 = t_3 = t_4 = 0$ and $H_0: r_1 = r_2 = r_3 = r_4 = 0$ at the level $\alpha = 0.05$.
2. If we are treating rows as the blocks, then the design is no more efficient than the corresponding randomized block design.

Estimation of a missing observation in an LSD

Let, in a $v \times v$ Latin square, the observation in the ith row and j^{th} column be missing. Let the missing observation be denoted by X, i.e. $y_{(ijk)} = X$. Let

- R = total of known observations in the ith row,
- C = total of known observations in the jth column,
- T = total of known observations for the kth treatment,
- S = total of known observations.

$$TSS = X^2 - \frac{2X(S+X)}{v^2} + \text{terms independent of } X$$

$$SS\ ROWS = \frac{(R+X)^2}{v} - \frac{(S+X)^2}{v^2} + \text{terms independent of } X$$

$$SS\ COLUMNS = \frac{(C+X)^2}{v} - \frac{(S+X)^2}{v^2} + \text{terms independent of } X$$

$$SS\ TMT = \frac{(T+X)^2}{v} - \frac{(S+X)^2}{v^2} + \text{terms independent of } X$$

$$SSE = TSS - (SS\ ROWS) - (SS\ COLUMNS) - (SS\ TMT)$$

$$= X^2 - \frac{2X(S+X)}{v^2} - \frac{(R+X)^2}{v} - \frac{(C+X)^2}{v} - \frac{(T+X)^2}{v} + \frac{3(S+X)^2}{v^2} + \text{terms independent of } X$$

We choose X which minimizes SSE. $\frac{\partial SSE}{\partial X} = 0$ gives

$$2X - \frac{2(S+X)}{v^2} - \frac{2(R+X)}{v} - \frac{2(C+X)}{v} - \frac{2(T+X)}{v} + \frac{6(S+X)}{v^2} = 0$$

$$\text{or} \qquad 2X\left(1 - \frac{4}{v^2} - \frac{3}{v}\right) = \frac{2}{v^2}(S - 4S) - \frac{2}{v}(R + C + T)$$

which simplifies to

$$X\left(v^2 - 4 - 3v\right) = v(R + C + T) - 2S$$

or

$$X = \frac{v(R + C + T) - 2S}{(v-1)(v-2)}$$

Note: after inserting the estimated value for the missing observation we carry out the ANOVA, subtracting one degree of freedom from TSS and consequently from SSE. The adjusted $SS(TMT)$ is obtained by subtracting

$$\frac{\left[(v-1)T + R + C - S\right]^2}{v(v-1)(v-2)}$$

from $SS(TMT)$.

Example

Estimate the missing value and set up the ANOVA for the LSD (use $\alpha = 0.01$).

A(2)	C(9)	B(1)	D(4)
C(8)	B(2)	D(6)	A(3)
B(4)	D(1)	A(–)	C(2)
D(2)	A(5)	C(7)	B(3)

Solution

Here $R = 7, C = 14, T = 10, S = 59$.

$$X = \frac{v(R + C + T) - 2S}{(v-1)(v-2)} = \frac{4(7 + 14 + 10) - 2 \times 59}{3 \times 2} = \frac{124 - 118}{6} = 1$$

The hypotheses to be tested are:

i. $H_0: r_1 = r_2 = r_3 = r_4$ against $H_a: r_i \neq r_{i'}$ for some $i \neq i'$

ii. $H_0: c_1 = c_2 = c_3 = c_4$ against $H_a: c_j \neq c_{j'}$ for some $j \neq j'$

iii. $H_0: t_1 = t_2 = t_3 = t_4 = 0$ against $H_a: t_k \neq t_{k'}$ for some $k \neq k'$

$$TSS = \sum_{i=1}^{4} \sum_{j=1}^{4} \left(y_{(ijk)}\right)^2 - \frac{G^2}{16} = 324 - 225 = 99$$

$$SS\ ROW = \sum_{i=1}^{4} \frac{R_i^2}{4} - \frac{G^2}{16} = \frac{970}{4} - \frac{60^2}{16} = 17.5$$

$$SS\ COL = \sum_{j=1}^{4} \frac{C_j^2}{4} - \frac{G^2}{16} = \frac{914}{4} - \frac{60^2}{16} = 3.5$$

$$SS\ TMT = \sum_{k=1}^{4} \frac{T_k^2}{4} - \frac{G^2}{16} = \frac{1066}{4} - \frac{60^2}{16} = 41.5$$

$$SS\ TMT_{(adj)} = 41.5 - \frac{\left[(v-1)T + R + C - S\right]^2}{v(v-1)(v-2)} = 41.5 - \frac{\left[(3 \times 10) + 7 + 14 - 59\right]^2}{4 \times 3 \times 2} = 39.72$$

$$SSE = TSS - (SS\ ROW) - (SS\ COL) - (SS\ TMT) = 99 - 17.5 - 3.5 - 41.5 = 36.5$$

Source of variation	d.f.	SS	MSS	F-ratio
Rows	3	17.5	5.83	$F_r = 0.8$
Columns	3	3.5	1.17	$F_c = 0.16$
Treatment (adj)	3	39.72	13.24	$F_t = 1.81$
Error	5	36.5	7.3	
Total	14	99		

$F_{0.01}[3, 5] = 12.1$. Since all the computed values of the test statistics are less than the tabular value, we accept all three null hypotheses.

Orthogonal Latin squares

Definition. Consider two Latin squares each of size $(s \times s)$. Superimpose one over the other. Two Latin squares are said to be **orthogonal** if each letter of the first square occurs with each letter of the second square once and only once.

Illustration. Consider two Latin squares L_1 and L_2:

L_1				L_2		
A	B	C		α	β	γ
B	C	A		β	γ	α
C	A	B		γ	α	β

Superimposing L_1 on L_2:

Aα	Bβ	Cγ
Bβ	Cγ	Aα
Cγ	Aα	Bβ

We note that L_1 and L_2 are orthogonal since they satisfy the requirement of orthogonality.

Identify A, B, C as treatments of one kind, say 3 maize varieties, and α, β, γ as treatments of another kind, say 3 different fertilizers. The design so obtained is called a **Graeco–Latin square design** (3×3).

Such orthogonal Latin squares are obtainable for p (strictly prime), say 3, 5, 7, 11, 13, 17, ..., or $p = s^n$ (a power of a prime), $n \in \mathbb{N}$, e.g. $2^2, 3^2, 4^2, 2^3, 3^3, ...$ However if $s = 4n + 2$, $n = 1, 2, ...$, then according to Euler no two orthogonal Latin squares exist. This conjecture has been disproved by R. C. Bose (1959) for $s = 10, 14, ...$, but for $s = 6$ it has been established that no pair of orthogonal (6×6) Latin squares exists; therefore $s = 6$ is called **Euler's number**.

Case I: p a strictly prime number

For $p = 5$ we can write 4 Latin squares (5×5) which are pairwise orthogonal.

L_1 **(drop 1 letter)**						L_2 **(drop 2 letters)**				
A	B	C	D	E		A	B	C	D	E
B	C	D	E	A		C	D	E	A	B
C	D	E	A	B		E	A	B	C	D
D	E	A	B	C		B	C	D	E	A
E	A	B	C	D		D	E	A	B	C

L_3 **(drop 3 letters)**						L_4 **(drop 4 letters)**				
A	B	C	D	E		A	B	C	D	E
D	E	A	B	C		E	A	B	C	D
B	C	D	E	A		D	E	A	B	C
E	A	B	C	D		C	D	E	A	B
C	D	E	A	B		B	C	D	E	A

L_1, L_2, L_3 and L_4 are pairwise orthogonal Latin squares. Identifying the letters with treatments and superimposing one square on another, we get a Graeco–Latin square (5×5) such as

Aα	Bβ	Cγ	Dδ	Eθ
Bβ	Cγ	Dδ	Eθ	Aα
Cγ	Dδ	Eθ	Aα	Bβ
Dδ	Eθ	Aα	Bβ	Cγ
Eθ	Aα	Bβ	Cγ	Dδ

A, B, C, D, E — maize varieties; $\alpha, \beta, \gamma, \delta, \theta$ — fertilizers.

CHAPTER 5: ANALYSIS OF GRAECO–LATIN SQUARES

Let $y_{(ijkl)}$ be the yield of treatment k of the first kind and treatment l of the second kind occurring in the plot (i, j). Assume

$$y_{(ijkl)} = \mu + r_i + c_j + t_k + \tau_l + e_{(ijkl)}$$
$$\text{var}\left(y_{(ijkl)}\right) = \sigma^2 \text{ (unknown constant)}$$

where

- μ is some unknown constant (usually the grand mean),
- r_i is a constant measuring the effect of the ith row,
- c_j is a constant measuring the effect of the jth column,
- t_k is a constant measuring the effect of the kth treatment of the first kind,
- τ_l is a constant measuring the effect of the lth treatment of the second kind,
- $e_{(ijkl)}$ is the random error, assumed to be $e_{(ijkl)} \sim N(0, \sigma_e^2)$.

We wish to test

i. $H_0: t_1 = t_2 = \cdots = t_s = 0$ against $H_a: t_k \neq t_{k'}$ for some $k \neq k'$
ii. $H_0: \tau_1 = \tau_2 = \cdots = \tau_s = 0$ against $H_a: \tau_l \neq \tau_{l'}$ for some $l \neq l'$
iii. $H_0: r_1 = r_2 = \cdots = r_s = 0$ against $H_a: r_i \neq r_{i'}$ for some $i \neq i'$
iv. $H_0: c_1 = c_2 = \cdots = c_s = 0$ against $H_a: c_j \neq c_{j'}$ for some $j \neq j'$

Step I. The method of least squares requires estimates of μ, r_i, c_j, t_k and τ_l such that

$$S = \sum_{i=1}^{v}\sum_{j=1}^{v}\left[y_{(ijkl)} - \left(\mu + r_i + c_j + t_k + \tau_l\right)\right]^2$$

is minimized.

Step II. Best estimators are:

$$\hat{\mu} = \bar{y}_{....} = \frac{G}{v^2} = \frac{1}{v^2}\sum_{i=1}^{v}\sum_{j=1}^{v} y_{(ijkl)}$$
$$\hat{r}_i = \bar{y}_{i...} - \bar{y}_{....}, \qquad \text{where } \bar{y}_{i...} = \frac{R_i}{v}$$
$$\hat{c}_j = \bar{y}_{.j..} - \bar{y}_{....}, \qquad \text{where } \bar{y}_{.j..} = \frac{C_j}{v}$$
$$\hat{t}_k = \bar{y}_{..k.} - \bar{y}_{....}, \qquad \text{where } \bar{y}_{..k.} = \frac{T_k}{v}, \quad T_k = \text{sum of observations of plots occupied by the } k^{\text{th}} \text{ treatment of the first kind}$$
$$\hat{\tau}_l = \bar{y}_{...l} - \bar{y}_{....}, \qquad \text{where } \bar{y}_{...l} = \frac{\Gamma_l}{v}, \quad \Gamma_l = \text{sum of observations of plots occupied by the } l^{\text{th}} \text{ treatment of the second kind}$$

By substituting these estimates in the model, rearranging, squaring both sides and summing over the rows and columns we can form the ANOVA table as follows.

ANOVA table

Source	df	SS	MSS	F-ratio
Rows	$v-1$	$\sum_i \sum_j \left(\bar{y}_{i...} - \bar{y}_{....}\right)^2$	$R = \frac{SS\ Rows}{v-1}$	$F_r = \frac{R}{E}$
Columns	$v-1$	$\sum_i \sum_j \left(\bar{y}_{.j..} - \bar{y}_{....}\right)^2$	$C = \frac{SS\ Columns}{v-1}$	$F_c = \frac{C}{E}$
Treatment (1st kind)	$v-1$	$\sum_i \sum_j \left(\bar{y}_{..k.} - \bar{y}_{....}\right)^2$	$T = \frac{SS\ TMT}{v-1}$	$F_t = \frac{T}{E}$
Treatment (2nd kind)	$v-1$	$\sum_i \sum_j \left(\bar{y}_{...l} - \bar{y}_{....}\right)^2$	$\Gamma = \frac{SS\ TMT(\Gamma)}{v-1}$	$F_\tau = \frac{\Gamma}{E}$
Error	$(v-1)(v-3)$	by subtraction	$E = \frac{SSE}{(v-1)(v-3)}$	
Total	v^2-1	$\sum_i \sum_j \left(y_{(ijkl)} - \bar{y}_{....}\right)^2$	-	

Tests for the above-stated hypotheses proceed by comparing the F-ratios with the tabular value at the specified level of significance.

CHAPTER 6: BALANCED INCOMPLETE BLOCK DESIGN (BIBD)

In most situations a randomized block design will be adequate. However, when the number of treatments exceeds the number of plots in each block and if the plot size is fixed, then not all treatments can be accommodated in each block. One way is to increase the size of the blocks, but this would introduce the problem of non-homogeneity within the block. But uniformity is an important requirement in the analysis of block designs. The only way out is to choose a subset of k treatments out of v treatments and assign them at random within a given block. Different subsets of treatments are assigned to different blocks. The resulting block design is known as an **incomplete block design** — incomplete because not all treatments occur in each block.

An incomplete block design is said to be **balanced** if every pair of treatments occur together the same number of times. That is, if, say, the treatments t_1 and t_2 occur together in 3 blocks, then every other pair of treatments must occur in 3 blocks for the design to be balanced.

We define a balanced incomplete block design as a block design in which

i. each treatment occurs once and only once in r blocks,

ii. each pair of treatments occur together in λ blocks.

Thus a BIBD is characterized by 5 parameters v, b, k, r and λ, where

- v = number of treatments,
- b = number of blocks,
- k = number of plots in each block,
- r = number of replications of each treatment,
- λ = number of times a pair of treatments occur together in the design.

These parameters satisfy the following conditions:

1. $bk = vr$
2. $\lambda(v-1) = r(k-1) \quad \Leftrightarrow \quad \lambda = \frac{r(k-1)}{v-1}$

Proof.

1. Since there are v treatments each replicated r times, the total number of plots in the design is vr. Further, since there are b blocks each of size k, there are bk plots in all. Hence $vr = bk$.
2. v treatments give rise to $\binom{v}{2}$ pairs, and since each pair occurs λ times, the total number of times all the pairs occur in the design is $\lambda\binom{v}{2}$.

But since the size of each block is k, each block gives rise to $\binom{k}{2}$ pairs; and since there are b blocks in all, the total number of treatment pairs in all the blocks is $b\binom{k}{2}$. Hence

$$\lambda\binom{v}{2} = b\binom{k}{2} \quad \Leftrightarrow \quad \lambda\frac{v(v-1)}{2} = b\frac{k(k-1)}{2} \quad \Leftrightarrow \quad \lambda v(v-1) = bk(k-1)$$

$$\Leftrightarrow \quad \frac{bk}{v}\lambda\,(v-1) = bk(k-1)$$

But $bk = vr$; hence

$$\lambda(v-1) = r(k-1).$$

The above two conditions are necessary but not sufficient for the existence of a BIBD.

Symmetric BIB design

A BIB design in which $v = b, r = k, \lambda$ arbitrary, is said to be a **symmetric design**, e.g. $v = b = 4, r = k = 3, \lambda = 2$:

Block	Treatments
Block 1	t_1, t_2, t_3
Block 2	t_2, t_3, t_4
Block 3	t_1, t_3, t_4
Block 4	t_1, t_2, t_4

This is a symmetric BIB design.

Methods of constructing BIB designs

I. Based on orthogonal sets of Latin squares

Consider orthogonal Latin squares (3×3):

L_1				L_2		
A	B	C		α	β	γ
B	C	A		β	γ	α
C	A	B		γ	α	β

Step I: Write down 9 numbers in a (3×3) square B:

1	2	3
4	5	6
7	8	9

Step II: Using the square B above, assign a block to the numbers in each row and column; we get six blocks:

Block	Numbers
Block 1	1, 2, 3
Block 2	4, 5, 6
Block 3	7, 8, 9
Block 4	1, 4, 7
Block 5	2, 5, 8
Block 6	3, 6, 9

Step III: Superimpose the square B on L_1 and assign blocks to the numbers which fall on A, B, C. We get three more blocks:

Block	Numbers
Block 7	1, 6, 8
Block 8	2, 4, 9
Block 9	3, 5, 7

Step IV: Repeat the procedure with L_2 to get three additional blocks:

Block	Numbers
Block 10	1, 5, 9
Block 11	2, 6, 7
Block 12	3, 4, 8

Step V: Identifying the numbers with treatments, we get a BIB design with parameters

$$v = 9, \quad b = 12, \quad r = 4, \quad k = 3, \quad \lambda = 1$$

Step VI: Take an additional block with the numbers 10, 11, 12 and 13, i.e.

$$10 \quad 11 \quad 12 \quad 13$$

Attach each one of the numbers to a set of 3 blocks. We get a new BIB design with parameters

$$v = 13, \quad b = 13, \quad r = 4, \quad k = 4, \quad \lambda = 1$$

This is a symmetric BIB design since $v = b, r = k$.

General method of constructing a BIB design

Step I: Start with pairwise orthogonal Latin squares of size s, where s is prime or a power of a prime. Let these be $L_1, L_2, \ldots, L_{s-1}$.

Step II: Write down s^2 numbers in a $(s \times s)$ square.

Step III: Assign blocks to the numbers in each row and each column. We get $2s$ blocks.

Step IV: Superimpose the square in Step II on $L_1, L_2, \ldots, L_{s-1}$. Assign blocks to the numbers which coincide with a fixed letter of $L_1, L_2, \ldots, L_{s-1}$; we get $(s-1)s$ additional blocks.

Step V: We get a total of $2s + s(s-1) = s^2 + s$ blocks, and identifying the s^2 numbers with treatments, we obtain a BIB design with parameters

$$v = s^2, \quad b = s^2 + s, \quad r = s + 1, \quad k = s, \quad \lambda = 1,$$

and the parameters satisfy the conditions

1. $bk = vr$
2. $\lambda(v-1) = r(k-1)$

We get a series of BIB designs for any power of a prime. Next take one additional block with $(s+1)$ treatments, namely

$$s^2+1, \quad s^2+2, \quad \ldots, \quad s^2+s+1,$$

and attach each of these treatments to a set of 's' blocks. We obtain another BIB design with parameters

$$v = s^2+s+1, \quad b = s^2+s+1, \quad r = s+1, \quad k = s+1, \quad \lambda = 1,$$

which is a symmetric BIB design.

Analysis of a BIB design

Let there be v treatments and b blocks. Let y_{ij} denote the observation of the ith treatment in the jth block ($i = 1, 2, \ldots, v$; $j = 1, 2, \ldots, b$). Assume

$$E\left(y_{ij}\right) = \mu + t_i + b_j, \qquad \text{var}\left(y_{ij}\right) = \sigma^2 \text{ a constant}, \qquad \text{cov}\left(y_{ij}, y_{i'j'}\right) = 0, \quad (i,j) \neq (i',j')$$

where

- μ is some fixed unknown constant,
- t_i is a constant measuring the effect of the ith treatment,
- b_j is a constant measuring the effect of the jth block,
- e_{ij} is the random error, assumed $N(0, \sigma^2)$.

We wish to test

$$H_0: t_1 = t_2 = \cdots = t_v = 0 \quad \text{against} \quad H_a: t_i \neq t_{i'} \text{ for some } i \neq i'.$$

Step I: Estimation of effects μ, t_i, b_j such that

$$S = \sum_{i=1}^{v} \sum_{j=1}^{b} \left[y_{ij} - \left(\mu + t_i + b_j\right)\right]^2$$

is minimized. The conditions of minimum are:

1. $\frac{\partial S}{\partial \mu} = 0$
2. $\frac{\partial S}{\partial t_i} = 0, i = 1, 2, \ldots, v$
3. $\frac{\partial S}{\partial b_j} = 0, j = 1, 2, \ldots, b$

Equations 1)–3) give rise to $(1 + v + b)$ equations. For uniqueness of solutions we assume

$$\sum_{i=1}^{v} t_i = 0, \qquad \sum_{j=1}^{b} b_j = 0.$$

Note that in this arrangement the estimates of t_i and b_j are correlated. The least-squares estimate of t_i is

$$\hat{t}_i = \frac{k\,Q_i}{v\lambda}$$

where

$$Q_i \text{ (adjusted treatment total)} = T_i - \frac{1}{k}\left(\text{sum of block totals in which } t_i \text{ occurs}\right)$$

that is

$$Q_i = T_i - \frac{1}{k} \sum_{j:\, t_i \in \text{ block } j} B_j, \qquad T_i = \sum_j y_{ij} = \text{sum of the plot yields in which } t_i \text{ occurs.}$$

Let

$$B_j = \sum_i y_{ij} = \text{sum of the plot yields which occur in block } j.$$

Then

$$SS\,TMT_{(adj)} = \sum_{i=1}^{v} \hat{t}_i Q_i = \frac{k}{v\lambda} \sum_{i=1}^{v} Q_i^2$$

$$SS\,BLOCK_{(unadj)} = \frac{1}{k} \sum_{j=1}^{b} B_j^2 - \frac{G^2}{vr}$$

$$TSS = \sum_{i=1}^{v} \sum_{j=1}^{b} y_{ij}^2 - \frac{G^2}{vr}$$

$$SSE = TSS - (SS\,TMT)_{adj} - (SS\,BLOCK)_{unadj}$$

We then form the ANOVA table.

ANOVA table I

Source of variation	d.f.	SS	MSS	F-ratio
Treatment (adj)	$v-1$	$(SS\ TMT)_{adj}$	T	$F=\frac{T}{E}$
Block (unadj)	$b-1$	$(SS\ BLOCK)_{unadj}$	*	*
Error	$vr-b-v+1$	by subtraction	E	
Total	$vr-1$	SST		

To test $H_0: t_1 = t_2 = \cdots = t_v$, compute $F = T/E$ and reject H_0 at level α whenever the computed

$$F > F_\alpha[\, v-1,\ vr-b-v+1\,] \text{ (tabular).}$$

If H_0 is rejected, recommend the treatment with maximum T_i.

A BIB is a non-orthogonal design in the sense that ($SS\ TMT$) and ($SS\ BLOCK$) are not additive. Hence the ANOVA table above is not suitable for testing $H_0: b_1 = b_2 = \cdots = b_b$. To be able to test for block effects we prepare Table II below.

ANOVA table II

Source of variation	d.f.	SS	MSS	F-ratio
Treatment (unadj)	$v-1$	$(SS\ TMT)_{unadj}$	*	*
Block (adj)	$b-1$	$(SS\ BLOCK)_{adj}$	B	$F=\frac{B}{E}$
Error	$vr-b-v+1$	by subtraction	E	
Total	$vr-1$			

To test $H_0: b_1 = b_2 = \cdots = b_b$, compute $F = B/E$ and reject H_0 at level α whenever the computed

$$F > F_\alpha[\, b-1,\ vr-b-v+1\,] \text{ (tabular).}$$

If H_0 is rejected then blocking is justified.

Example

An experiment was conducted to compare the effect of $v = 7$ chemical substances on the skin of male rats. The area of experimentation on an animal's skin was confined to a region which was known to be relatively homogeneous, but this restricted the experimental units (patches of skin) per animal. Hence, to eliminate the rat-to-rat variability in the comparison of treatments, the experiment was blocked on the rats using the BIBD shown below ($k = 3, r = 3, b = v = 7, \lambda = 1$). The seven blocks correspond to the seven rats.

Block			
B1	t_1 10.2	t_4 12.9	t_3 11.7
B2	t_5 9.1	t_2 8.8	t_5 9.2
B3	t_1 11.3	t_2 6.9	t_6 14.1
B4	t_2 12.1	t_7 7.7	t_7 8.6
B5	t_6 15.2	t_3 9.7	t_4 14.2
B6	t_3 9.9	t_5 8.6	t_4 14.3
B7	t_6 10.3	t_1 13.1	t_7 6.2

Do the data present sufficient evidence to indicate a real difference in the effect of the chemical substances on the skin? (use $\alpha = 0.05$)

Solution

Let $n_{ij} = 1$ if the ith treatment is in the jth block and zero otherwise.

	1	2	3	4	5	6	7
B_j	31.3	36.9	32.4	31.1	33.7	37.5	27.2
T_i	34.6	27.8	31.3	41.4	26.9	45.6	22.5
$\sum_j n_{ij}B_j$	96.0	97.4	96.5	99.3	101.0	108.1	92.0
Q_i	2.6	−4.66	−0.86	8.3	−6.9	9.6	−8.3

$$G = 230.1, \qquad \frac{G^2}{vr} = \frac{(230.1)^2}{21} = 2521.238$$

$$TSS = \sum_{i=1}^{v}\sum_{j=1}^{b} y_{ij}^2 - \frac{G^2}{vr} = 2684.97 - 2521.238 = 163.732$$

$$SS\ BLOCK_{(unadj)} = \frac{1}{k}\sum_{j=1}^{7} B_j^2 - \frac{G^2}{21} = \frac{7640.05}{3} - 2521.238 = 25.445$$

$$SS\ TMT_{(adj)} = \frac{k}{v\lambda}\sum_{i=1}^{7} Q_i^2 = \frac{3}{7}(302.18) = 129.506$$

$$SSE = TSS - (SS\ TMT)_{adj} - (SS\ BLOCK)_{unadj} = 163.732 - 129.506 - 25.445 = 8.78$$

ANOVA table I

Source of variation	d.f.	SS	MSS	F-ratio
Treatment (adj)	6	129.56	21.584	$F = 21.584/1.098 = 19.66$
Block (unadj)	6	25.445	*	*
Error	8	8.78	1.098	
Total	20	163.732		

$F = 19.66 > F_{0.05}[6, 8] = 3.58$. Hence reject H_0 and conclude that there is evidence that the treatment effects are different.

To test for block effects prepare Table II and use the fact that

$$(SS\ TMT)_{adj} + (SS\ BLOCK)_{unadj} = (SS\ TMT)_{unadj} + (SS\ BLOCK)_{adj}$$

(the source prints the error sum of squares in this table as 8.28; the computed value above is 8.78, and $8.78/8 = 1.098$ *is the mean square used for the F-ratio.)*

CHAPTER 7: COMPARISON OF DESIGNS

Suppose we are interested in carrying out a field experiment to test the hypothesis of equality of treatment effects. Given a choice between (1) CRD, (2) RBD and (3) LSD, which field experiment would you prefer? In this chapter we outline procedures for answering the above question.

Criterion for selection

We choose a design whose efficiency is maximum.

Definition. The observation in each field has a variance σ^2. The efficiency of a design is

$$\varepsilon = \frac{1}{\sigma^2}.$$

For example:

1. For the CRD the model is

$$y_{ij} = \mu + t_i + e_{ij}, \qquad e_{ij} \sim N\left(0, \sigma^2_{CRD}\right); \quad i = 1, 2, ..., v;\ j = 1, 2, ..., n$$

so that

$$\text{eff}_{CRD} = \frac{1}{\text{var}\left(y_{ij}\right)_{CRD}} = \frac{1}{\sigma^2_{CRD}}$$

The estimated efficiency of the CRD is

$$\text{est. eff} = \frac{1}{\widehat{\sigma}^2_{CRD}} = \frac{1}{SE}$$

2. For the equivalent RBD the model is

$$y_{ij} = \mu + t_i + b_j + e_{ij}, \qquad i = 1, 2, ..., v;\ j = 1, 2, ..., b; \quad e_{ij} \sim N\left[0, \sigma^2_{RBD}\right]$$

so that

$$\text{est. eff} = \frac{1}{\widehat{\sigma}^2_{RBD}} = \frac{1}{SE_{RBD}}$$

I. Comparison of designs — fixed effects models

Comparison of CRD and RBD

We display the ANOVA tables side by side as follows.

ANOVA table for the RBD — fixed effects model

Source of variation	df	SS	MSS	$E(SS)$
Treatment	$v-1$	$(SS\,TMT)_{RBD}$		$(v-1)\sigma^2_{RBD} + b\sum_i t_i^2$
Block	$b-1$	$(SS\,BLOCK)_{RBD}$		$(b-1)\sigma^2_{RBD} + v\sum_j b_j^2$
Error	$(v-1)(b-1)$	SSE_{RBD}	E_{RBD}	$(v-1)(b-1)\sigma^2_{RBD}$
Total	$vb-1$	TSS	–	

ANOVA table for the equivalent CRD — fixed effects model

Source of variation	df	SS	MSS	$E(SS)$
Treatment	$v-1$	$(SS\,TMT)_{CRD}$		$(v-1)\sigma^2_{CRD} + b\sum_i t_i^2$
Error	$v(b-1)$	SSE_{CRD}	E_{CRD}	$v(b-1)\sigma^2_{CRD}$
Total	$bv-1$	TSS	–	

The total sum of squares in the RBD equals the total sum of squares in the CRD, so that

$$E\{TSS_{RBD}\} = E\{TSS_{CRD}\}.$$

That is,

$$(b-1)\sigma^2_{RBD} + v\sum_j b_j^2 + (v-1)(b-1)\sigma^2_{RBD} = (bv-1)\sigma^2_{CRD}$$

which gives

$$\sigma^2_{CRD} = \frac{(b-1)\sigma^2_{RBD} + v\sum_j b_j^2/(v-1)}{(bv-1)}$$

so that, replacing the components by their estimates,

$$\widehat{\sigma}^2_{CRD} = \frac{(b-1)E_{RBD} + b(v-1)B}{(bv-1)}$$

where B is the block mean square and E_{RBD} the error mean square of the RBD. Hence

$$\text{Estimated value of efficiency of CRD} = \frac{1}{\widehat{\sigma}^2_{CRD}} = \frac{bv-1}{(b-1)E_{RBD} + b(v-1)B}$$

$$\text{Estimated efficiency of RBD} = \frac{1}{\widehat{\sigma}^2_{RBD}} = \frac{1}{E_{RBD}}$$

and therefore the **relative efficiency of the RBD to the equivalent CRD** is

$$R.E. = \frac{\text{eff}_{RBD}}{\text{eff}_{CRD}} = \frac{\widehat{\sigma}^2_{CRD}}{\widehat{\sigma}^2_{RBD}} = \frac{(b-1)E_{RBD} + b(v-1)B}{(bv-1)E_{RBD}}$$

> *[The algebra of this comparison occupies source pp. 58–60; the identities used are those of the two ANOVA tables above.]*

CASE I: Suppose $(SS\ BLOCK)$ is significant, i.e. $F_B = B/E_{RBD}$ exceeds the tabular value, so that $H_0: b_1 = \cdots = b_b$ is rejected. Then

$$\frac{(b-1)E_{RBD} + b(v-1)B}{(bv-1)E_{RBD}} > 1,$$

i.e. the block effects are not all equal and the RBD has higher efficiency than the equivalent CRD.

CASE II: When $H_0: b_1 = b_2 = \cdots = b_b$ is true, then $F_B = 1$ and $R.E. = 1$, that is, the RBD is equally efficient as the equivalent CRD.

Comparison of the $(s \times s)$ Latin square design with the RBD

The model for the LSD is

$$y_{(ijk)} = \mu + r_i + c_j + t_k + e_{(ijk)}, \qquad e_{(ijk)} \sim N\left(0, \sigma^2_{LSD}\right)$$

ANOVA table for the LSD (with $E(SS)$ shown):

Source	df	SS	$E(SS)$
Rows	$s-1$	$\sum_i \sum_j \left(\bar{y}_{i..} - \bar{y}_{...}\right)^2$	$(s-1)\sigma^2_{LSD} + s\sum_i r_i^2$
Columns	$s-1$	$\sum_i \sum_j \left(\bar{y}_{.j.} - \bar{y}_{...}\right)^2$	$(s-1)\sigma^2_{LSD} + s\sum_j c_j^2$
Treatment	$s-1$	$\sum_i \sum_j \left(\bar{y}_{..k} - \bar{y}_{...}\right)^2$	$(s-1)\sigma^2_{LSD} + s\sum_k t_k^2$
Error	$(s-1)(s-2)$	$\sum_i \sum_j \left(y_{(ijk)} - \bar{y}_{i..} - \bar{y}_{.j.} - \bar{y}_{..k} + 2\bar{y}_{...}\right)^2$	$(s-1)(s-2)\sigma^2_{LSD}$
Total	s^2-1	$\sum_i \sum_j \left(y_{(ijk)} - \bar{y}_{...}\right)^2$	

The model for the equivalent RBD is

$$y_{(ik)} = \mu + r_i + t_k + e_{ik}, \qquad e_{ik} \sim N\left(0, \sigma^2_{RBD}\right)$$

ANOVA table for the equivalent RBD with rows as blocks:

Source of variation	df	SS	$E(SS)$
Treatment	$s-1$	$\sum_i \sum_k \left(\bar{y}_{i.} - \bar{y}_{..}\right)^2$	$(s-1)\sigma^2_{RBD} + s\sum_k t_k^2$
Block	$s-1$	$\sum_i \sum_j \left(\bar{y}_{.j} - \bar{y}_{..}\right)^2$	$(s-1)\sigma^2_{RBD} + s\sum_i r_i^2$
Error	$(s-1)(s-1)$	$\sum_i \sum_j \left(y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..}\right)^2$	$(s-1)(s-1)\sigma^2_{RBD}$
Total	s^2-1	$\sum_i \sum_j \left(y_{ij} - \bar{y}_{..}\right)^2$	

The total sum of squares for the LSD equals the total sum of squares for the RBD, so

$$E\{\text{Total SS for LSD}\} = E\{\text{Total SS for RBD}\},$$

which yields, after substituting the components and solving for σ^2_{RBD},

$$\widehat{\sigma}^2_{RBD} = \frac{(s-1)E_{LSD} + (s-1)C}{(s-1)^2}$$

where C is the column mean square of the LSD and E_{LSD} its error mean square, so that

$$\text{Estimated efficiency of LSD} = \frac{1}{E_{LSD}}, \qquad \text{Estimated efficiency of RBD} = \frac{1}{\widehat{\sigma}^2_{RBD}}$$

and the **relative efficiency of the LSD to the equivalent RBD** is

$$R.E. = \frac{E_{RBD}}{E_{LSD}} = \frac{(s-1)E_{LSD} + (s-1)C}{(s-1)E_{LSD}}$$

From the LSD ANOVA we can test significant effects of rows, columns and treatments via

$$F_1 = \frac{R}{E_{LSD}}, \qquad F_2 = \frac{C}{E_{LSD}}, \qquad F_3 = \frac{T}{E_{LSD}}$$

CASE I: If $H_0: c_1 = c_2 = \cdots = c_s$ is false, then $F_2 = C/E_{LSD} > 1$ and

$$R.E. = \frac{(s-1)E_{LSD} + (s-1)C}{(s-1)E_{LSD}} > 1.$$

Conclusion: if the s columns are unequal or non-homogeneous, then the LSD is more efficient than the corresponding RBD.

CASE II: If $H_0: c_1 = c_2 = \cdots = c_s$ is true then $F_2 = C/E_{LSD} = 1$ and $R.E. = 1$.

Conclusion: if the columns are homogeneous, then the LSD and the RBD are equally efficient.

Comparison of the LSD and the CRD

The model for the equivalent CRD is

$$y_{ij} = \mu + t_i + e_{ij}, \qquad e_{ij} \sim N\left(0, \sigma^2_{CRD}\right)$$

ANOVA table for the equivalent CRD — fixed effects model:

Source of variation	df	SS	$E(SS)$
Treatment	$s-1$	$(SS\ TMT)$	$(s-1)\sigma^2_{CRD} + s\sum_i t_i^2$
Error	$s(s-1)$	SSE	$s(s-1)\sigma^2_{CRD}$
Total	s^2-1	TSS	

The total sum of squares in the CRD equals the total sum of squares in the LSD, so

$$E\{TSS_{CRD}\} = E\{TSS_{LSD}\},$$

that is

$$(s-1)\sigma^2_{CRD} + s\sum_i t_i^2 + s(s-1)\sigma^2_{CRD} = (s-1)\sigma^2_{LSD} + s\sum_i r_i^2 + (s-1)\sigma^2_{LSD} + s\sum_j c_j^2 + (s-1)\sigma^2_{LSD} + s\sum_k t_k^2 + (s-1)(s-1)\sigma^2_{LSD}$$

Hence

$$\widehat{\sigma}^2_{CRD} = \frac{(s-1)E_{LSD} + (s-1)R + (s-1)C}{(s-1)^2},$$

where R and C are the row and column mean squares of the LSD, so that

$$\text{Estimated efficiency of CRD} = \frac{1}{\widehat{\sigma}^2_{CRD}}, \qquad \text{Estimated efficiency of LSD} = \frac{1}{E_{LSD}}$$

and the relative efficiency of the LSD to the equivalent CRD is obtained from these two estimated efficiencies.

> *[The algebra of the comparisons in this chapter runs from source p. 57 to p. 64, where the notes end.]*